Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · LangGraph Multi-Agent · MCP Servers · RAG Pipelines · LLM Evaluation · Agentic Workflows · Production AI · Python · AWS

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PROFESSIONAL SUMMARY

Builder FDE with 8 years shipping production AI systems and enterprise backend infrastructure. Built **Leasel**: an 8-agent LangGraph multi-agent system (Triage → Retrieval → Reasoning → HITL Escalation) with production-grade RAG pipelines, hybrid search, citation grounding, and durable state management - 93.6% precision across 1,240 compliance documents, hierarchical delegation with conditional routing (exact match for Google JD preferred quals: LangGraph + ReAct + multi-agent + HITL). Built **AgentProbe** (open-source): evaluation pipelines and observability framework for agentic systems - 5-metric eval engine (correctness, groundedness, latency, **token efficiency**, **hallucination rate**), CI/CD regression harness, golden datasets, React observability dashboard with per-run drift and latency tracking, LangSmith tracing integration (exact match for JD responsibilities: "build evaluation pipelines and observability frameworks for agentic systems"). At Entrata: **7 years owning enterprise-scale production systems** - 100K+ MAU, 180K domain records, 99.5% uptime, zero compliance violations. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026 - preferred qual met).

TECHNICAL SKILLS

Languages & Frameworks: Python (expert — 8yr), TypeScript, JavaScript, Java, Kotlin, SQL (expert — 200+ production optimizations), PL/SQL, PHP, R, Bash; FastAPI, Spring Boot, Django, Node.js, React.js; REST APIs, GraphQL, OpenAPI

Agentic AI & LLM: LangGraph (multi-agent StateGraph, durable state, conditional routing), LangChain, Multi-Agent Systems, Agent Orchestration, Hierarchical Delegation, ReAct Pattern, MCP (Model Context Protocol), MCP Servers, Agentic Workflows, Tool Calling, HITL (Human-in-the-Loop), RAG Pipelines, RAG Architectures, Hybrid Search, Semantic Chunking, Citation Grounding, Vector Databases (ChromaDB, Qdrant), LLM Evaluation, LLM-native Metrics (token efficiency, latency, cost-per-request), Observability Frameworks, Evaluation Pipelines, Agent Tracing, LangSmith, State Management, Granular Tracing, LLM-as-Judge, Eval-Driven Development, Regression Harnesses, Golden Datasets, Hallucination Detection, Guardrails, Prompt Injection Defense, OpenAI API, Anthropic API, Groq, Ollama, AWS Bedrock, Vertex AI, Gemini Models (familiar), Google AI APIs (familiar), Enterprise AI Deployment

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, SageMaker, Step Functions, API Gateway, IAM, SQS, OpenSearch), GCP / Google Cloud Platform (Cloud Storage, Vertex AI — familiar), Kubernetes (K8s), Docker, Apache Kafka, RabbitMQ, Redis, CI/CD (GitHub Actions), Terraform, Microservices, Distributed Systems, Fault-Tolerant Systems, PostgreSQL, Snowflake, Apache Airflow, Celery, Prometheus, Grafana, OpenTelemetry, Mapbox (geospatial APIs — familiar)

ML & Data: PyTorch, scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, Feature Engineering, Statistical Modeling, NumPy, Pandas, Tableau, Power BI, MLflow, HuggingFace, BERT, spaCy, ETL/ELT Pipelines, Data Governance, Data Lineage

PROJECTS

Leasel: Agentic AI Platform — 8-Agent LangGraph, Multi-Agent, HITL, RAG

MULTI-AGENT RAG PIPELINE

Python · LangGraph (8-agent StateGraph) · FastAPI · ChromaDB · TF-IDF · Hybrid Search · OCR · SQLite — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an **8-agent LangGraph multi-agent system** with durable state and hierarchical delegation: Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → **Human Escalation (HITL)**. Conditional routing based on confidence state — ReAct-adjacent pattern; enforced design rule in code: LLM extracts and reasons; deterministic logic makes the final call — auditability by architecture. (Exact match: preferred quals — LangGraph + multi-agent systems + ReAct + hierarchical delegation + HITL.)
- **Production-grade RAG pipeline**: OCR-fallback ingestion, section-aware chunking, **hybrid search** (TF-IDF + semantic reranking), citation grounding to exact source clauses, missing/conflicting clause detection, conditional HITL escalation on low confidence. Achieved **93.6% precision, 87.9% F1** across 10 compliance rules on 1,240 residential lease PDFs (5 US states + federal); pass@1 81.7%, pass@3 88.3% via AgentProbe regression harness. Demonstrates enterprise AI deployment from conception to production.

AgentProbe: Evaluation Pipelines & Observability Framework for Agentic Systems

OPEN SOURCE EVAL & OBSERVABILITY

Python · LangGraph · FastAPI · ChromaDB · Docker · GitHub Actions · React · LangSmith · OpenAI · Anthropic · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Built **evaluation pipelines and observability framework** for agentic systems (exact JD responsibility): **5-metric eval engine** with LLM-native metrics — correctness, groundedness, latency, **token efficiency**, **hallucination rate**; CI-integrated **regression harness with golden datasets**; multi-provider LLM routing (OpenAI/Anthropic/Groq/Ollama); offline + online evaluation loops; zero vendor dependency.
- 3-agent LangGraph runtime (Orchestrator → RAG Tracer → Auto Fix) with tool calling and inter-agent state handoffs; React **observability dashboard** tracks per-run metrics, retrieval drift, and latency in real time; **granular tracing via LangSmith**; deployed as open-source — reusable field pattern extracted into module (exact JD responsibility: "convert repeatable field patterns into reusable modules"). Evaluated Leasel across 1,240+ compliance test cases with CI/CD eval gates on every commit.

Magizhs: Agentic Mission Planning — FDE Builder Pattern (Prototype → Production)

FDE PATTERN RAG + GUARDRAILS

Python · LangGraph · RAG · FastAPI · FAA Part 107 Corpus — Magizhs Tech Solution Internship 2025

- FDE builder pattern: stakeholder request → rapid prototype → production agentic workflow → operational deployment. Built a **LangGraph stateful agentic system** (parse → plan → retrieve → validate → conditional routing) converting natural-language survey requests (marketing, real estate, commercial) into structured flight plans; RAG layer over FAA Part 107 corpus with cited compliance checks; **validation guardrails** enforcing regulatory limits — non-compliant missions blocked with grounded rejection reasons before MAVLink/QGroundControl export. Demonstrates rapid prototype to production agentic workflow delivery alongside domain stakeholders (exact JD responsibility).

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 – Present

AI/ML Engineer (Volunteer) — ML Models · Data Pipelines · AI Integration

- Developing ML models and data analysis pipelines for software applications and BI projects; integrating AI components with back-end and front-end teams; building pipelines that surface insights from raw business data into application-facing APIs.

George Mason University Aug 2024 – Aug 2025

Research Associate & Graduate Teaching Assistant — ML Modeling · Anomaly Detection · Analytics Engineering

- Built multi-source prediction pipelines (Python, R, 190+ countries) achieving **R^2 = 0.977**; applied Isolation Forest for automated data integrity across 3 research pipelines; fine-tuned custom classification model (HuggingFace, PyTorch) with inference API on Quart; designed Tableau/ggplot2 dashboards driving **40% improvement** in research comprehension.
- Delivered technical workshops to **600+ students** across multiple sections (Python, OOP, SQL, database design); provided one-on-one debugging support and graded assignments with rubric-aligned feedback.

Entrata — PropTech / Real Estate SaaS Sep 2018 – Aug 2024

Senior Software Engineer — Enterprise Production Systems · Agentic AI Pipelines · AWS · Compliance at Scale

- **Owned the HUD income-limit compliance pipeline for 7 years** — federal source rows expanded into ~180,000 domain records powering income eligibility, rent calculations, HAP payment processing, and tax-credit compliance for 100K+ housing units. Every release included baseline validation, reconciliation, and rollback preparation. **Zero compliance violations across entire ownership period. 99.5% uptime.**
- Built and owned **RabbitMQ-based async processing system** for gross-rent updates across thousands of tenant records — idempotency guarantees and retry logic eliminating double-processing failures; owned through production incidents and annual rent-schedule changes. **Optimized 200+ SQL queries (30% performance gain)** via execution plan analysis, indexing, partitioning; complex multi-tenant production schemas.
- Achieved **100% PHI compliance** on AWS S3 document migration with Libsodium-encrypted pipelines (zero security incidents); led microservices Docker redesign (**50% scalability improvement, 30% deployment reduction**); eliminated 80% manual operational workload via Python/Bash automation; **300+ code reviews, 5+ engineers mentored**. Shipped 20+ full-stack production modules across **120+ Agile sprints**.

Entrata India Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · Data Pipelines · AWS S3 · Compliance Workflows

- Built full-stack property management modules (PHP, PostgreSQL, JavaScript); designed SQL import pipelines for 1.6M HUD/Tax Credit income limit records; developed compliance workflows and AWS S3 document integrations; participated in sprint planning, code reviews, and performance debugging.

EDUCATION

M.S. Data Analytics Engineering — George Mason University May 2026 · GPA 3.87

AI pipelines, ML systems, statistical modeling, data engineering, agentic AI, LLM evaluation, anomaly detection, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur 2013 – 2017